FinRobot: audit before investment use

Columbia STAT GR5398 | Fall 2026 | Assignment 1 | Evidence cutoff: October 11, 2026

Findings and investment judgment

The actual finrobot_equity two-step pipeline generated eight typed-agent sections and HTML reports for NVDA, AMD, INTC, AAPL and MSFT, all verified as Technology by FMP profiles. GOOGL was also run to complete the course starter set. Each company has at least two peer inputs. This is broad sector coverage: semiconductor, hardware and software peers have limited valuation comparability.

Would I hand these reports to a portfolio manager? Only as clearly labelled research drafts with an audit attached. News access returned HTTP 402, yet the news agent asserted headlines and analyst consensus. Regulatory, competitive and supply-chain assertions often lacked supplied evidence. A successful model response is not evidence of factual correctness.

Two-company financial audit: 20 NVDA/AMD metric comparisons produced 16 matches and four EBITDA or EBITDA-margin definition differences. Original issuer filings and SEC facts are preserved. AMD cash-flow D&A is $750m plus $2,254m amortization, or $3,004m. The FMP EBITDA definition differs from operating income plus D&A; full non-operating reconciliation remains unresolved.

The forecast patch fixes historical years mislabelled as forecasts: future-year validity rose from 8/15 to 15/15 in actual FMP outputs. However, 68 supplemental rolling-origin SEC tests show pooled MAPE worsening from 13.7372% to 15.8580%. The improvement is temporal consistency, not demonstrated predictive accuracy.

Five DCF recomputations agree arithmetically. Valuation quality remains weak: CFO less capex is not unlevered FCFF, net debt is synthetically set at 10% of EV, and the share denominator is FY weighted-average diluted shares. Growth, WACC and terminal growth are assumptions. No defensible Buy/Sell recommendation follows from these mechanical scenarios.

Course deadline is October 12, 2026. The exact hour and submission timezone could not be established; the CourseWorks timezone America/New_York does not prove the GitHub submission cutoff. No midnight assumption, publication or submission was made.

Company financial evidence and valuation scenarios

Company / period end	Revenue $bn	FCF $bn	DCF $/share
NVDA 2026-01-25	215.94	96.68	71.66
AMD 2025-12-27	34.64	6.74	74.80
INTC 2025-12-27	52.85	-4.95	0.00
AAPL 2025-09-27	416.16	98.77	119.60
MSFT 2026-06-30	331.84	66.99	163.31

Revenue and FCF are annual amounts in USD, not live TTM measures. Source: preserved stable income/cash-flow CSVs in actual_runs/baseline and audit/fmp_coverage.csv. DCF source: audit/fmp_valuation_audit.csv. Intel raw negative valuation is floored at zero by the helper; zero is not a fair-value conclusion.

NVDA shows strong growth and margins but baseline forecast-label errors and a high valuation narrative needing verification. AMD revenue growth is accompanied by EBITDA-definition caveats. Intel negative FCF makes a simple constant-growth DCF unsuitable. Apple and Microsoft positive FCF supports cash generation, not an attractive entry price without defensible capital-cost and growth assumptions.

Assumptions and sensitivity

Upstream DCF uses ten projection years, growth 10% for years 1-5 and 5% for years 6-10, WACC 10%, terminal growth 2.5%, and synthetic net debt 10% of EV. Independent full recalculation spans WACC 8%, 10%, 12% and terminal growth 1.5%, 2.5%, 3.5% for all five companies: 45 saved scenarios. Forecast patch uses historical CAGR capped at +/-50% and flat margins. These are transparent heuristic assumptions.

Runtime and cost

Actual initial two-step company runtimes range from 35.5 to 42.0 seconds, excluding installation, audit calls and rendering updates. Saved-response cost subtotal is $1.0731, based on 203,218 input and 87,917 output tokens. Rates: $2 input, $0.50 cached input, $8 output per million tokens. Several diagnostic calls and one failed/truncated evaluation response are excluded, so this is not a billing reconciliation.

Grounding and repeated-input stability

Section	Claims	Supported	Rate
company_overview	105	14	13.3%
competitor_analysis	73	8	11.0%
investment_overview	81	44	54.3%
major_takeaways	63	33	52.4%
news_summary	62	18	29.0%
risks	104	22	21.2%
tagline	24	5	20.8%
valuation_overview	64	33	51.6%

Three companies, all eight sections: 576 extracted claims. Classification totals: {'Unsupported': 397, 'Supported': 177, 'Contradicted': 2}. These are provisional automated rates using the same model for extraction and classification. Extraction completeness and entailment were not independently adjudicated. Nonliteral table citations are flagged separately; formatting mismatch alone does not invalidate support.

Five documented review corrections were applied. Examples: historical 2025A revenue was misrepresented as forward guidance; a compound claim combined correct revenue numbers with an unsupported customer-adoption explanation. Original model classifications and corrected claim tables remain available. Lowest supported rates occur in competitor and company overview sections, consistent with prompts asking for contextual facts absent from financial tables.

NVDA was generated three times with identical saved user prompts and unchanged agent instructions. The extracted top-three category labels were Market Risk, Competitive Risk and Operational Risk in all three runs, while supporting wording and specifics differed. Label agreement is 3/3 categories across runs, but the original system prompt specifies this category order. Thus label agreement is largely structural; risk_text_stability.csv measures differing wording. It does not prove factual accuracy. No explicit rating or numeric target price was emitted in the agent prose, so their variances are undefined, not zero. Step 2 market-fetched or default header fields are separate outputs.

Upstream header limitations include assumed 95% free float and trailing P/E labelled forward P/E. Raw report HTML retains the baseline for auditing, with a warning banner. Missing FMP news is disclosed; the prompts have no implemented web-search tool despite instructions referring to online searches.

Requirements, reproducibility and remaining steps

Course requirement	Status / evidence
Specified code, starter five and five-company sector universe	Executed; actual_execution_summary.csv and sector_evidence.csv
Two companies, ten financial metrics	Audited; 16 matches, four definition discrepancies disclosed
Forecast / valuation and sensitivity	Executed; independent DCF plus 45 scenarios
Eight-section grounding across three companies	Provisional automated audit plus five corrections; independent complete review remains
Same inputs three times	Executed; ratings/targets absent, category-label risks stable
One improvement with before / after	Patch applied; temporal validity improved, MAPE worsened
Research report / Medium article	Research artifacts prepared; Medium outline is unpublished
Name_UNI folder and course submission PR	Not submitted; student identity not supplied and user prohibits push/submission

Environment: separate Python 3.12.14 virtual environment, requirements.lock.txt. Original finrobot_equity commit: 2717499b8e30f242640af08c4ad9afd1113c2d45. Three clean-applying patches: forecast improvement, legacy FMP-to-stable transport, and typed-agent wiring/evidence recording. Provider failures raise rather than creating canned success text. Tracing is disabled and credentials are excluded.

Reproduction commands and exact source-file paths are in CURRENT_STATUS.md. Run run_data_only.py, analyze_fmp_actual.py, run_actual_pipeline.py and evaluate_agents.py with the private config path; review_grounding.py applies recorded corrections. Final Notebook executes the audits. Cached financial requests and frozen prompts avoid unnecessary repeat costs. Source dates and retrieval times remain attached to raw data.

Sources: course/FinRobot_README.md and the course GitHub Assignment1/FinRobot directory; FMP stable endpoints; SEC CompanyFacts; original NVIDIA annual report PDF and AMD issuer-hosted 10K. Filing URLs and metric lineage are recorded in audit/filing_crosscheck.csv and audit/sec_fact_lineage.csv. The program-wide research goals were not added as separate Assignment 1 requirements.

Personal final steps: review the evidence, conclusions and AI-assistance disclosure; resolve remaining financial-definition and grounding questions; write/publish your own Medium analysis if permitted; fill Name_UNI and follow the course submission instructions. No GitHub push, email, course submission or publication was performed. FinRL and its Positron kernel were untouched.